

MATH120-24S2 Discrete Mathematics

Assignment 6

1. (a) A spy intercepts the following message that was encrypted by the standard Caesar cipher:

PD WH PDKL ND RUD

What was the message? Show your working.

- (b) Alison (Alice's absent-minded cousin) was using an affine cipher with alphabet $\mathbb{Z}_{26}$ but left her cheat sheet lingering around:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
10	17	24	05	12	19	00	07	14	21	02	09	16	23	04	11	18	25	06	13	20	01	08	15	22	03
K	R	Y	F	M	T	A	H	O	V	C	J	Q	X	E	L	S	Z	G	N	U	B	I	P	W	D

- i. Determine a, b in the encryption function $e(x) = ax + b$.
- ii. Determine the decryption function d : write $d(y) = a'x + b'$ for some $0 \leq a', b' \leq 25$.

Solution:

- (a) We replace every letter by the letter that came 3 letters earlier in the alphabet: MA TE MAHI KA ORA *Through hard works comes fulfilment.*

- (b) i. We have that 00 gets mapped to 10. Since $a \cdot 0 + b = 10$, we see that $b = 10$. Now 01 is mapped to 17, so $a \cdot 1 + 10 = 17$. It follows that $a = 7$. So $e(x) = 7x + 10$.
- ii. Since we have an explicit table, we can easily deduce the decryption function in the same way as we have found the encryption function. We see that $d(0) = 6$ and $d(1) = 21$. We find that $d(y) = 15y + 6$.

Without using the table, we could retrieve the decryption function from the encryption function with the following general method:

The decryption function is given by $(y - 10) \cdot 7^{-1}$ so we need to find the inverse of 7 mod 26. Now

$$26 = 3 \cdot 7 + 5$$

$$7 = 1 \cdot 5 + 2$$

$$5 = 2 \cdot 2 + 1$$

So

$$\begin{aligned} 1 &= 5 - 2 \cdot 2 \\ &= 5 - 2 \cdot (7 - 1 \cdot 5) \\ &= 3 \cdot 5 - 2 \cdot 7 \\ &= 3 \cdot (26 - 3 \cdot 7) - 2 \cdot 7 \\ &= 3 \cdot 26 - 11 \cdot 7 \end{aligned}$$

So $-11 = 15$ is the inverse of 7 in $\mathbb{Z}_{26}$. The decryption function becomes $d(y) = 15(y - 10) = 15y - 150 = 15y + 6$ since $-150 \equiv 6 \pmod{26}$.

2. Alice and Bob use the RSA Crypto System to send messages. Alice chooses $p = 19$, $q = 29$ and publishes her public key $(n = 551, e = 295)$

- (a) Verify that $e = 295$ is a valid choice for the RSA algorithm.
- (b) We asked ChatGPT to compute d , the private key of Alice. After an initial wrong guess, we received the following:

We're computing d using e and $\phi(n)$, where $e = 295$ and $\phi(n) = 504$.

We want to find d such that $de \equiv 1 \pmod{\phi(n)}$.

So, we need to solve for d in the equation:

$$295d \equiv 1 \pmod{504}$$

To find d , we can use the Extended Euclidean Algorithm or use Euler's theorem.

Let's find d using the Extended Euclidean Algorithm. We start with the equation:

$$504 = 295 \cdot 1 + 209$$

Then,

$$295 = 209 \cdot 1 + 86$$

$$209 = 86 \cdot 2 + 37$$

$$86 = 37 \cdot 2 + 12$$

$$37 = 12 \cdot 3 + 1$$

Now, working backward:

$$1 = 37 - 12 \cdot 3$$

$$1 = 37 - (86 - 37 \cdot 2) \cdot 3 = -86 \cdot 3 + 37 \cdot 7$$

$$1 = -86 \cdot 3 + (209 - 86 \cdot 2) \cdot 7 = 209 \cdot 7 - 86 \cdot 17$$

$$1 = 209 \cdot 7 - (295 - 209 \cdot 1) \cdot 17 = -295 \cdot 17 + 209 \cdot 24$$

$$1 = -295 \cdot 17 + (504 - 295 \cdot 1) \cdot 24 = 504 \cdot 24 - 295 \cdot 41$$

So, we have found that $d = 41$.

Therefore, $d = 41$, not $d = 407$. Thank you for catching that mistake!

Once again, ChatGPT is wrong. What is the correct answer? (The reasoning above could actually be useful).

- (c) Bob wants to send the message $m = 32$ to Alice. What does he send? *Hint:* $2^{466} \equiv 100 \pmod{551}$.

Solution:

- (a) Since $p = 19, q = 29, \phi(n) = 18 \cdot 28 = 504$. We have to check that $e = 295$ is invertible in $\mathbb{Z}_{\phi(n)} = \mathbb{Z}_{504}$, this is true if $\gcd(295, 504) = 1$. Now $295 = 5 \cdot 59$ and $504 = 2^3 \cdot 3^2 \cdot 7$, so there are no common prime factors, so indeed, $\gcd(295, 504) = 1$.
- (b) The calculations using the Euclidean algorithm are correct, and we have that, indeed, $1 = 504 \cdot 24 - 295 \cdot 41$. This means that $-295 \cdot 41 \equiv 1 \pmod{504}$. Since $-41 \equiv 463 \pmod{504}$, 463 is the inverse of 295 in $\mathbb{Z}_{504}$. Thus $d = 463$.
- (c) Bob sends $(32)^{295} \pmod{551} \equiv (2^5)^{295} \pmod{551}$. Hence, we find:

$$\begin{aligned} (2^5)^{295} \pmod{551} &\equiv 2^{5 \cdot 295} \pmod{551} \\ &\equiv 2^{2 \cdot 504 + 466 + 1} \pmod{551} \\ &\equiv (2^{504})^2 \cdot 2^{466} \cdot 2 \pmod{551} \\ &\equiv 1^7 \cdot 100 \cdot 2 \pmod{551} \\ &\equiv 200 \pmod{551} \end{aligned}$$

Here we have used the hint and that $2^{504} \equiv 1 \pmod{551}$ (Euler's theorem).

3. Alice and Bob are using the RSA cryptosystem with public key

$$(n = 61771, e = 5).$$

Eavesdropper Eve has found out that $\phi(n) = 61272$. Use $\phi(n)$ to find the primes p and q that were used to construct $n = pq$ in the public key. *Do not use brute force to solve this problem. Your solution should use a method that would work for much larger numbers too.*

Solution:

We have that $n = pq = 61771$ and that $\phi(n) = (p - 1)(q - 1) = pq - (p + q) + 1 = 61272$. Hence,

$$pq = 61771$$

and

$$p + q = 61771 - 61272 + 1 = 500.$$

Then p and q are the solutions of

$$X^2 - 500X + 61771.$$

which are given by

$$\frac{500 \pm \sqrt{500^2 - 4 \cdot 61771}}{2} = \frac{500 \pm 54}{2}.$$

Hence, the values for p and q are 277 and 223.